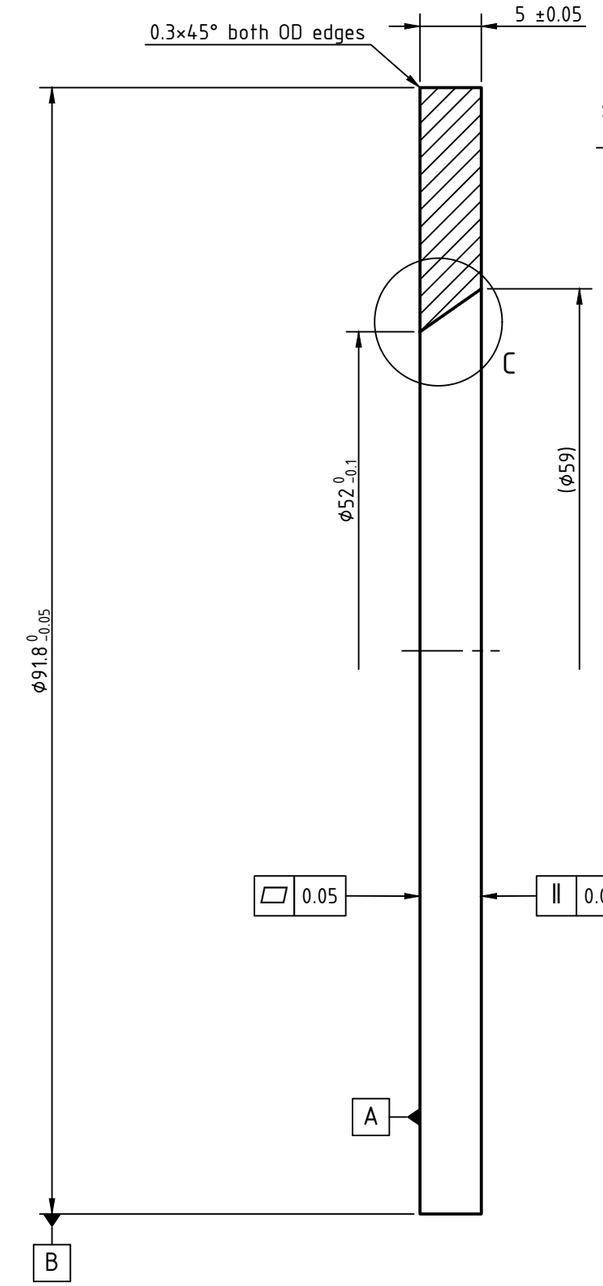
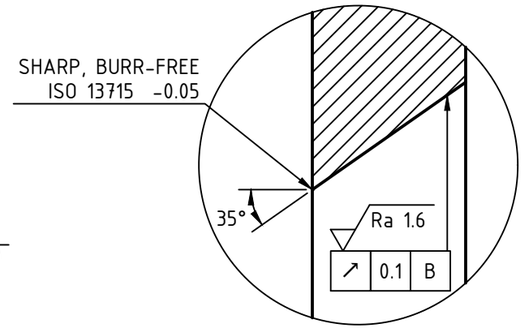
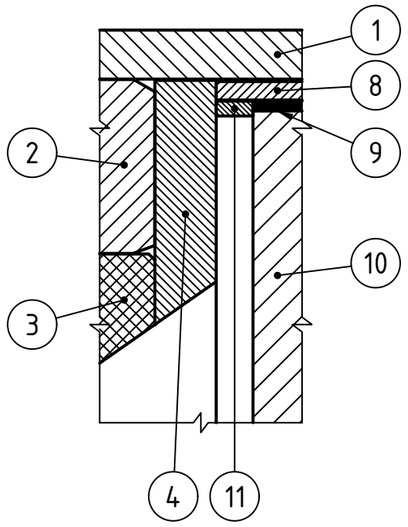




DETAIL C (5:1)

INSTALLATION (REFERENCE) 2:1



NOTES

1. Material: glass-fabric reinforced phenolic laminate, NEMA LI 1 grade G-3, or compression-moulded glass-phenolic (MXB-360 / Durite class); density about 1.85 g/cm³. G-10 / G-11 are epoxy-glass grades and are not acceptable. Certificate of conformity.
2. Stock: sheet 6.35 (1/4 in) min, faced both sides to 5; laminations parallel to the faces. Turn OD and bore in one set-up on a mandrel located on datum A.
3. Machine dry with carbide or PCD tools, with dust extraction. No coolant, no oil. Blow clean with dry, oil-free air; deliver and store bagged, clean and dry.
4. The bore edge on the aft face is sharp, free of burrs and chips (detail C): it meets the graphite entry ϕ52 of insert SRM100-104 with no forward-facing step. ϕ52 is never exceeded.
5. Edges: OD edges 0.3×45° (not in the 3-D model): the part slides through the honed seal bore of case SRM100-101 at assembly. Other edges per the title block.
6. Loose fit in the aft honed zone of case SRM100-101, bore ϕ92.000 to 92.10 (A +0.030/0, A ≤ 92.070, SRM100-101 note 4): diametral clearance 0.20 to 0.35. Not bonded; located radially by the case bore. Aft face (A) down on the holder and graphite forward faces; clamped axially by the aft grain spacer on its forward face (grain stack, take-up ring). Liner SRM100-110 also rests on its forward face and carries no axial load. Grease the OD and edges: Parker Super O-Lube or DC-55, thin film.
7. Inspect 100 %: thickness at 4 places 90° apart, OD, bore ϕ52 at the aft face (optical comparator or taper gauge), flatness and parallelism on a surface plate, runout on a mandrel. Visual: no delamination, cracks or chipped edges. Record values.
8. Mark SRM100-105 and serial on the aft face between ϕ70 and ϕ86, indelible ink or laser, no engraving.
9. Single use: a new part for every firing, scrapped after it (charred bore; README, After every firing). Never re-use a fired part.
10. Used on SRM100-000 item 4 (aft closure bench subassembly). Item numbers in the installation view per SRM100-000.

Ra 3.2 (✓)

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Glass phenolic NEMA LI 1 G-3	38.8 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE INSULATOR Phenolic washer on the holder face - bore continues the 35° convergent

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-105	A	1 / 1